

04.02 Fractional Exponents

Practice problems

Show your work and state restrictions or excluded values when needed.

Recognize	Apply these rules
An exponent has a denominator.	Use the denominator as the root index.
An exponent has a numerator.	Raise the base to that power.
The problem mixes powers and roots.	Rewrite in the form that makes simplification easiest, then apply exponent rules.

A. Rewrite as radicals

1.

$$x^{1/2}$$

2.

$$x^{3/2}$$

3.

$$a^{2/3}$$

4.

$$y^{5/4}$$

5.

$$m^{7/3}$$

6.

$$p^{3/5}$$

7.

$$b^{1/6}$$

8.

$$z^{4/7}$$

9.

$$q^{5/2}$$

10.

$$r^{2/5}$$

11.

$$t^{7/2}$$

12.

$$u^{3/8}$$

13.

$$v^{5/6}$$

14.

$$w^{9/4}$$

15.

$$s^{2/7}$$

16.

$$c^{11/3}$$

17.

$$d^{4/9}$$

18.

$$n^{5/8}$$

19.

$$k^{3/10}$$

20.

$$h^{7/5}$$

B. Evaluate

1.

$$16^{3/4}$$

2.

$$27^{2/3}$$

3.

$$81^{1/2}$$

4.

$$32^{3/5}$$

5.

$$64^{2/3}$$

6.

$$125^{2/3}$$

7.

$$16^{5/4}$$

8.

$$243^{2/5}$$

9.

$$25^{3/2}$$

10.

$$256^{3/4}$$

11.

$$729^{1/3}$$

12.

$$625^{1/4}$$

13.

$$49^{3/2}$$

14.

$$8^{4/3}$$

15.

$$36^{3/2}$$

16.

$$1000^{2/3}$$

17.

$$121^{3/2}$$

18.

$$1024^{3/5}$$

19.

$$64^{5/6}$$

20.

$$128^{4/7}$$

C. Simplify using exponent rules

1.

$$x^{2/3} x^{1/3}$$

2.

$$a^{5/4} a^{3/4}$$

3.

$$y^{7/6} / y^{1/6}$$

4.

$$m^{9/5} / m^{4/5}$$

5.

$$(x^{1/2})^4$$

6.

$$(a^{2/3})^3$$

7.

$$(y^{3/4})^2$$

8.

$$p^{1/2}p^{3/2}$$

9.

$$b^{5/6}/b^{1/3}$$

10.

$$(q^{2/5})^5$$

11.

$$r^{7/8} r^{1/8}$$

12.

$$s^{11/6} / s^{5/6}$$

13.

$$(t^{3/7})^7$$

14.

$$u^{4/5} u^{1/5}$$

15.

$$(v^{1/3})^6$$

16.

$$w^{9/4} / w^{1/4}$$

17.

$$(c^{2/9})^9$$

18.

$$d^{3/8}d^{5/8}$$

19.

$$(n^{4/5})^5$$

20.

$$k^{7/10}/k^{2/10}$$